METRIC
MIL-DTL-62415B(AT)
30 September 1998
SUPERSEDING
MIL-PRF-62415A(AT)
11 March 1996

DETAIL SPECIFICATION

PUMP, SUBMERSIBLE, BILGE, ELECTRIC-MOTOR-DRIVEN

This specification is approved for use by the US Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 24 volt electric-motor-driven, submersible bilge pump for moving water (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4320

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts (See 4.3.1).
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of (See 4.3.1).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

ARMY

12369000	- Chemical Agent Resistant Coatings (CARC) Paint Systems Index.
12292104	- Pump Unit, Submersible.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117 - Standard Practice for Operation of Salt Spray (Fog) Apparatus (DoD Adopted).

(Application for copies may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design and construction. The pump shall be designed and constructed as specified in Drawing 12292104, and shall be in accordance with 3.2.1 and 3.2.1.1.

3.2.1 Electric motor. The pump shall have an enclosed waterproof electric motor which shall be rated at 28 volts direct current (Vdc).

3.2.1.1 Bearings. The motor shall have anti-friction, lubricated for life, sealed bearings, rated for continuous duty.

3.3 Materials. Materials used in the fabrication of the pump shall be as specified herein and in referenced drawings. When not thereby specified, the materials shall be in accordance with the manufacturer's specifications for submersible pumps, as long as those materials are capable of meeting all the operating, support and ownership, and environmental requirements as specified herein.

3.3.1 Hazardous materials. Asbestos and cadmium materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcuries per gram or activity per item equals or exceeds 0.01 microcuries.

3.3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.4 Operating requirements.

3.4.1 Overall performance. When operated at 28 ± 0.05 Vdc at any indicated head pressure, the pump output (flow) and current consumption shall be as specified in figure 1.

3.4.2 Water level. The pump shall start to deliver water when the depth of water is not more than 7.5 centimeters (cm) (3 inches (in.)), above the pump mounting surface. The pump may stop delivering water when the depth of water is not more than 7 cm (2.75 in.), above the pump mounting surface.

3.4.3 Pumping capacity. The pump shall deliver not less than 125 liters per minute (L/m) (33 gallon per minute (gal/min)) against a 2.3 meter (m) (7.5 foot (ft)) static pressure head through a 3.55 ± 0.15 cm (1.40 ± 0.06 in.) inside diameter outlet.

3.4.4 Dry-run capability. After 25 hours of dry running, the pump shall retain not less than 90 percent (%) of the pumping capacity as specified in 3.4.3.

3.4.5 Air-lock limitation. The pump shall not air-lock for more than 20 seconds under any pumping condition.

3.4.6 Dielectric strength. The pump shall withstand exposure to a 500 V root-mean-square (rms), 60 hertz (Hz) potential for 1 minute without evidence of puncture in solid insulation, continuing sparkover, breakdown of the dielectric material, or damage to the insulation sufficient to cause failure of the motor.

3.4.7 Starting capacity. The motor of the pump shall start with 13.5 to 15 Vdc applied at –1 degree Celsius (°C) (30 degrees Fahrenheit (°F)).

3.4.8 Electromagnetic interference (EMI). The pump shall meet the EMI requirements as specified in figures 2 and 3.

3.5 Interface requirements.

3.5.1 Interchangeability. All assemblies and components having the same part number shall be functionally and dimensionally interchangeable.

3.5.2 Weight. The weight of the pump shall be no greater than 5.4 kilograms (kg) (11.9 pounds (lb)).

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3.6 Support and ownership requirements.

3.6.1 Corrosion protection. Corrosion protection shall be as specified in 3.6.1.1 through 3.6.1.3.

3.6.1.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided.

3.6.1.2 Steel components. Steel components shall withstand at least 2 hours of a 5 % salt solution spray without evidence of corrosion.

3.6.1.3 Aluminum components. Aluminum components shall withstand at least 336 hours of a 5 % salt solution spray without evidence of corrosion.

3.6.2 Endurance. After being operated for 100 hours, the pump shall retain not less than 90% of the pumping capacity as specified in 3.4.3.

3.6.3 Finish. Unless otherwise specified (see 6.2), exposed ferrous and aluminum surfaces shall be cleaned, pretreated, primed, and painted in accordance with an applicable chemical agent resistant coating (CARC) paint system selected from Drawing 12369000 for external application. The color of paint final topcoat shall be green 383.

3.6.3.1 Hydrogen embrittlement. Ferrous metal parts with a Rockwell hardness of C40 or greater shall be baked for hydrogen embrittlement relief for parts receiving pickle cleaning or electroplating operations within four hours of cleaning and/or electroplating operation. Such parts shall be heat treated after phosphating for 8 hours at 99 to 107°C (210 to 225°F) or shall be held for 240 hours at room temperature to relieve any embrittlement due to hydrogen. Parts having a hardness of RC48 or higher shall not be subjected to phosphating.

3.6.4 Identification marking. Unless otherwise specified (see 6.2), a nameplate shall be attached to the pump housing in the manner specified on Drawing 12292104 and shall include the following:

a. “Bilge pump”
b. “28 Vdc”
c. Manufacturer’s identification
d. Serial number
e. Military part number
f. Federal stock number
g. Manufacturer’s part number
h. Date of manufacture
i. Contract number
j. “US”

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3.7 Operating environment requirements.

3.7.1 High temperature. Subsequent to attaining a temperature of 71°C (185 °F), the pump shall meet the overall performance requirements as specified in 3.4.1 at an ambient temperature of 65 ± 3°C (149 ± 5°F).

3.7.2 Low temperature. Subsequent to attaining a temperature of -54 ± 3°C (-65 ± 5°F), and at an ambient temperature of 2 ± 2°C (36 ± 4°F), the pump shall meet the overall performance requirements as specified in 3.4.1.

3.7.3 Shock. The pump shall withstand sawtooth waveform shock pulses and durations as specified in figure 4 along each of the three major axes without deforming, cracking, or breaking, and without compromising the pumping capacity as specified in 3.4.3.

3.7.4 Vibration. The pump shall withstand vibration levels and durations as specified in figures 5 and 6 in each of the three major axes without deforming, cracking, or breaking, and without compromising the pumping capacity as specified in 3.4.3.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
    1. Preproduction inspection (see 4.1.1.1).
    2. Initial production inspection (see 4.1.1.2).
b. Conformance inspections (see 4.1.2).

4.1.1 First article inspection. When a first article sample is required (see 3.1), first article inspection shall be performed on samples as specified in 4.1.1.1 and 4.1.1.2. Unless otherwise specified (see 6.2), first article inspection shall include all of the verifications as specified in table I, further elaborated as follows:

a. The examinations shall be as specified in 4.3.2.
b. The materials and construction verifications shall be as specified in 4.3.3. In order to verify that steel and aluminum materials are corrosion resistant, each sample shall also be subjected to and pass the tests as specified in 4.3.5.1 and 4.3.5.2.
c. The operating, support and ownership, and operating environment tests shall be as specified in 4.3.4.1 through 4.3.6.4, in the order as specified in table II.

4.1.1.1 Preproduction inspection. When specified (see 6.2 and 6.3), a preproduction sample shall consist of three pumps (samples A, B, and C).

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4.1.1.2 Initial production inspection. Unless otherwise specified (see 6.2 and 6.3), the Government shall select three pumps (samples A, B, and C) from the first lot produced under the production contract for initial inspection.

Table I. Verification methods.

Title	Require-ments	Verific-ation	Title	Require-ments	Verific-ation
Design and construction	3.2	4.3.2 & 4.3.3	**Support and ownership requirements**	3.6	4.3.5
Electric motor	3.2.1	4.3.3	Dissimilar metals	3.6.1.1	4.3.3
Bearings	3.2.1.1	4.3.3	Steel components	3.6.1.2	4.3.5.1
Materials	3.3	4.3.3	Aluminum components	3.6.1.3	4.3.5.2
Hazardous materials	3.3.1	4.3.3	Endurance	3.6.2	4.3.5.3
Operating requirements	3.4	4.3.4	Finish	3.6.3	4.3.2 & 4.3.3
Overall performance	3.4.1	4.3.4.1			
Water level	3.4.2	4.3.4.2	Hydrogen embrittlement	3.6.3.1	4.3.2 & 4.3.3
Pumping capacity	3.4.3	4.3.4.3			
Dry-run capability	3.4.4	4.3.4.4	Identification marking	3.6.4	4.3.2
Air-lock limitation	3.4.5	4.3.4.5	**Operating environment requirements**	3.7	4.3.6
Dielectric strength	3.4.6	4.3.4.6			
Starting capacity	3.4.7	4.3.4.7	High temperature	3.7.1	4.3.6.1
EMI	3.4.8	4.3.4.8	Low temperature	3.7.2	4.3.6.2
Interface requirements	3.5	4.3.2	Shock	3.7.3	4.3.6.3
Interchangeability	3.5.1	4.3.2	Vibration	3.7.4	4.3.6.4
Weight	3.5.2	4.3.2			

Table II. Order of first article testing.

Sample	Title	Verification
A	Pumping capacity	4.3.4.3
	Endurance	4.3.5.3
	Air-lock limitation	4.3.4.5
	Shock	4.3.6.3
	Vibration	4.3.6.4
B	Overall performance	4.3.4.1
	Water level	4.3.4.2
	Pumping capacity	4.3.4.3
	Air-lock limitation	4.3.4.5
	Dielectric strength	4.3.4.6

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Table II. Order of first article testing – Continued.

Sample	Title	Verification
C	Pumping capacity	4.3.4.3
	Dry-run capability	4.3.4.4
	High temperature	4.3.6.1
	Low temperature	4.3.6.2
	Air-lock limitation	4.3.4.5
	Shock	4.3.6.3
	Vibration	4.3.6.4
	Starting capability	4.3.4.7
	EMI	4.3.4.8

4.1.2 Conformance inspections. Unless otherwise specified (see 6.2 and 6.4), conformance inspection samples, designated as group A and group B, shall be selected from an inspection lot (see 6.5) in accordance with the contractual sampling plan, and shall undergo the verifications as specified in 4.1.2.1 and 4.1.2.2.

4.1.2.1 Group A samples. Group A samples shall be examined as specified in 4.3.2 and shall be subjected to and pass the dielectric strength and the starting capability tests as specified in 4.3.4.6 and 4.3.4.7.

4.1.2.2 Group B samples. Group B samples shall be examined as specified in 4.3.2 and shall be subjected to and pass the pumping capacity test as specified in 4.3.4.3.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following standard (room) ambient conditions:

a. Air temperature: 25 ± 6°C (77 ± 10°F).
b. Relative humidity: (uncontrolled room ambient).
c. Atmospheric pressure: (site pressure).

4.2.1 Temperature stabilization. Test specimens shall be stabilized at room temperature for 1 hour before testing.

4.2.2 Test set-up. Unless otherwise specified herein, the pump shall be positioned with the axes of rotation horizontal during testing to simulate in-use operation.

4.2.3 Environmental tests. When specified herein, the pump shall be operated at ambient conditions prior to proceeding with any of the environmental tests to obtain a baseline for determining satisfactory operation of the item during and after the tests, as applicable.

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4.3 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously approved or previously qualified designs.

4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedure, to verify performance.

4.3.2 Examinations. Each pump selected for first article and conformance inspections shall undergo visual and equipmental examinations for the following:

a. Design and construction shall be in accordance with Drawing 12292104.
b. Assemblies and components shall be functionally and dimensionally interchangeable.
c. External application finish shall be in accordance with Drawing 12369000.
d. The pump weight shall be no more than 5.4 kg (11.9 lb).
e. Identification marking shall be in accordance with Drawing 12292104 and shall include the following: “Bilge pump”, “28 Vdc”, “U.S.”, the manufacturer’s identification and part number, the serial number, the military part number, the federal stock number, the date of manufacture, and the contract number.
f. Throughout the testing procedures as specified in 4.3.4.1 and 4.3.6.4, ferrous metal parts baked for hydrogen embrittlement relief, shall show no evidence of embrittlement.

4.3.3 Material, design, and construction verification. With the exception of material corrosion resistance testing (see 4.3.5.1 and 4.3.5.2), the manufacturer’s standard practices shall be used to verify that pump materials, design, and construction conform to the requirements of this specification. Conformance shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.3.4 Operating requirements tests.

4.3.4.1 Overall performance. The pump shall be operated against a static pressure head up to 6.4 m (20.9 ft) at 28 $\pm$ 0.05 Vdc. The output volume and current consumption shall be as specified in figure 1.

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4.3.4.2 Water level. The pump shall be operated when submerged and the water level shall be allowed to drop until the water discharge stops. With the pump running, raise the water level and observe the water level relative to the pump mounting surface at which the pump begins to discharge water. The pump shall start to deliver water when the depth of water is not more than 7.5 cm (3 in.), above the pump mounting surface. Allow the water level to drop and observe the level at which the water discharge stops. The pump may stop delivering water when the depth of water is not more than 7 cm (2.75 in.), above the pump mounting surface.

4.3.4.3 Pumping capacity. Under the following conditions, the pump shall deliver not less than 125 L/m (33 gal/min) through a 3.55 ± 0.15 cm (1.40 ± 0.06 in.) inside diameter outlet:

a. The static pressure head shall be 2.3 m (7.5 ft).
b. The voltage at motor terminals shall be 28 ± 0.05 Vdc.
c. The ambient air temperature shall be 25 ± 6°C (77 ± 11°F).
d. The specific gravity of water being pumped shall be no greater than 1.03.
e. The water temperature shall be 25 ± 6°C (77 ± 11°F).

4.3.4.4 Dry-run capability. The pump shall be dry-run for 25 hours under the following conditions:

a. The voltage at motor terminals shall be 26 ± 0.05 Vdc.
b. The ambient air temperature shall be 65 ± 6°C (149 ± 10.8°F).
c. The pump shall be attached to a discharge line as defined in figure 7.
d. There shall be no maintenance or adjustments to the pump permitted during the test.

After the dry-run the pumping capacity shall be determined in accordance in 4.3.4.3. The pump shall deliver not less than 113 L/m (30 gal/min).

4.3.4.5 Air-lock limitation. Occurrence of air-lock, if any, shall be observed and shall not last for more than 20 seconds.

4.3.4.6 Dielectric strength. The pump shall be tested in accordance with the method 301 of MIL-STD-202, or equivalent (see 4.3.1), where the test voltage shall be 500 V rms at 60 Hz and applied for 1 minute between ground and each winding or electrical circuit element. The voltage source shall be capable of delivering 1 milliampere (mA) to a 500 ohm resistive load. Brushes shall be raised from commutator and winding shall be ungrounded. The filter capacitor in the electric connector shall be disconnected during this test. After the test, there shall be no malfunction, evidence of evidence of puncture in the solid insulation, continuing sparkover, breakdown of the dielectric material, or any damage to the insulation sufficient to cause failure of the motor.

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4.3.4.7 Starting capability. Apply a voltage in the range of 13.5 to 15 Vdc, applied at -1 °C (30 °F), to the pump electrical terminals. The pump shall start.

4.3.4.8 EMI test. The pump shall be tested in accordance with MIL-STD-462, or equivalent (see 4.3.1), for the EMI limits as specified in figures 2 and 3.

4.3.5 Support and ownership requirements verifications.

4.3.5.1 Steel corrosion resistance test. Actual pump components shall be subjected to the salt spray test specified by ASTM B117 for a minimum of 2 hours using a 5% salt solution. Prior to this test, the samples shall be immersed in a solution of nitric acid and sodium dichromate to dissolve surface contaminants resulting from fabrication processes such as machining. There shall be no evidence of corrosion.

4.3.5.2 Aluminum corrosion resistance test. Corrosion resistance of aluminum shall be determined on separate test panels 0.64 to 1.60 millimeters (mm) (0.025 to 0.63 in.) thick and not less than 76 x 254 mm (3 x 10 in.) in width and length. The panels shall be subjected to no less than 336 hours of salt spray corrosion testing in accordance with ASTM B117, except that significant surface shall be inclined six degrees from vertical. There shall be no more than 15 scattered spots or pits, none larger than 0.8 mm (0.03 in.) diameter, on a total of 968 square centimeters (cm^2) (150 square inches (in^2)). There shall be no more than 5 scattered spots or pits, none larger than 0.8 mm (0.03 in.) diameter, on a total of 194 cm^2 (30 in^2) of test area from one or more test pieces.

4.3.5.3 Endurance. The pump shall pump water containing sodium chloride, sand and clay mixed in the following proportions and under the following conditions:

a. Mix 95 parts (by volume) fresh water and 5 parts sodium chloride.
b. For each 189 liters (L) (50 gallon) of solution in a, add 2.3 kg (5 lb) of clay (Bentonite) and 0.9 kg (2 lb) of sand. The sand shall be 0.01 to 1.0 mm (0.0004 - 0.04 in.) in diameter with the majority of grains 0.15 to 0.30 mm (0.006 - 0.012 in.) in diameter.
c. Circulation and agitation shall be such that the sand and clay remain in suspension throughout the test.
d. The test conditions shall be as specified in 4.3.4.3.

4.3.5.3.1 Procedure. The endurance test shall be performed as follows:

a. Water, as specified in 4.3.5.3, shall be pumped for 6 hours at 2.3 m (7.5 ft) water head and a flow of not less than 125 L/m (33 gal/min).
b. The pump shall rest for not less than 30 minutes.
c. Three cycles shall be performed as follows:

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1. Three hours of pumping, as specified in step a.
2. Twenty minutes of dry-run.
3. A rest of not less than 30 minutes.
4. Three hours of pumping, as specified in step a.
5. A rest of not less than 30 minutes.

d. The above steps (a through c) shall be performed four times. After the test, the pump capacity shall be determined in accordance with 4.3.4.3. The pump shall deliver not less than 113 L/m (30 gal/min).

4.3.6. Operating environment requirements tests.

4.3.6.1 High temperature. The pump shall be tested as specified in 4.3.4.3 a, b, and d, except that the pump and the water shall attain a temperature of 71°F (185°F), and then shall be stabilized at 65 ± 3°C (149 ± 5°F). At the stabilized temperature, the pump shall be subjected to and pass the test as specified in 4.3.4.3, a, b, and d.

4.3.6.2 Low temperature. The pump shall be tested as specified in 4.3.4.3 a, b, and d, except that the pump and the water shall attain a temperature of -54 ± 3°C (-65 ± 5°F). The pump and water shall then be stabilized at 2 ± 2°C (36 ± 4°F) and shall be subjected to and pass the test as specified in 4.3.4.3, a, b, and d.

4.3.6.3 Shock. The pump shall be mounted in a position to simulate its actual use and three shocks in each direction shall be applied along its three mutually perpendicular axis (total of 18 shocks). The shock pulse shape and duration shall be as specified in figure 4. After the test, the pump shall be subjected to and pass the test as specified in 4.3.4.3.

4.3.6.4 Vibration. The pump shall be mounted in a manner that will simulate its actual use, and shall be vibrated along each axis in accordance with the vibration levels specified in figure 5. The sweep time shall be 15 minutes for the sweep frequency range of 5 to 500 to 5 Hz. The cycling time shall not exceed 3 hours in each axis. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 6. The specified sweep time is that of an ascending plus a descending sweep and is twice the ascending sweep time shown on figure 6 for the specified range. When the pump resonances below 5 Hz are measured, the test curve shall be extended to 2 Hz and the sweep time shall be 18 minutes (2 to 500 to 2 Hz). There shall be no operational tests required during vibration. There shall be no evidence of deformation, cracking, or breaking. After the test, the pump shall be subjected to and pass the test as specified in 4.3.4.3.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The pump covered by this specification is intended to expel water and contaminants (sea water, fuels, and lubrications) from the hulls of combat vehicles. The pump unit is military unique because it must meet with all the military CARC coating procedures described in the referenced Army Drawing 12369000.

6.2 Acquisition requirements. Acquisition documents must supply the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.2.2, and 2.3).
c. If first article inspection is required (see 3.1).
d. If finish should be other than specified (see 3.6.3).
e. If identification marking should be other than as specified (see 3.6.4).
f. If first article verifications are other than as specified (see 4.1.1).
g. If preproduction test is specified (see 4.1.1.1).
h. If initial production inspection is other than as specified (see 4.1.1.2).
i. If conformance inspection is other than as specified (see 4.1.2).
j. Specification of conformance inspection sampling plan (see 4.1.2).
k. Packaging requirements (see 5.1).

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously

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acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes applied, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table II and their designated frequency based on a risk assessment for the procurement.

6.5 Lot formation. An inspection lot consists of all the pumps from an identifiable production period, form one manufacturer, submitted at one time for inspection.

6.6 Subject term (key word) listing.

Draw fluid
Horizontal flow
Output flow
Water
Waterproof

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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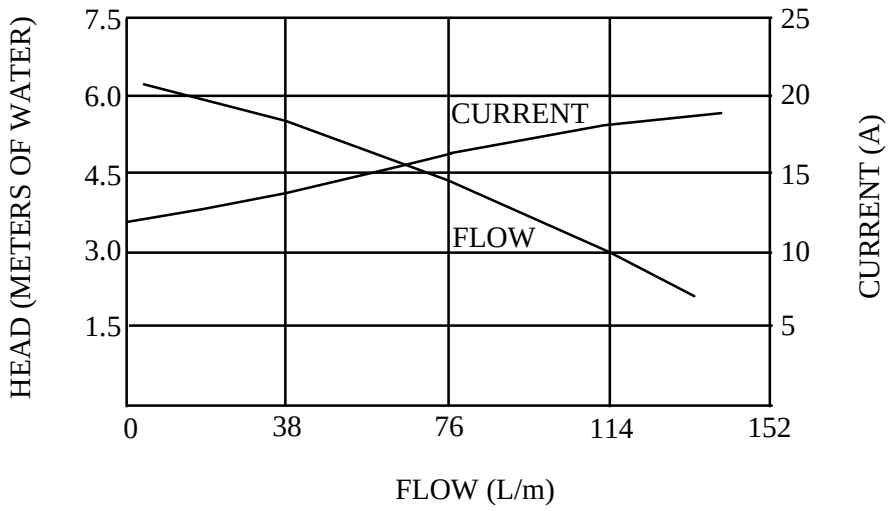


FIGURE 1. Performance data.

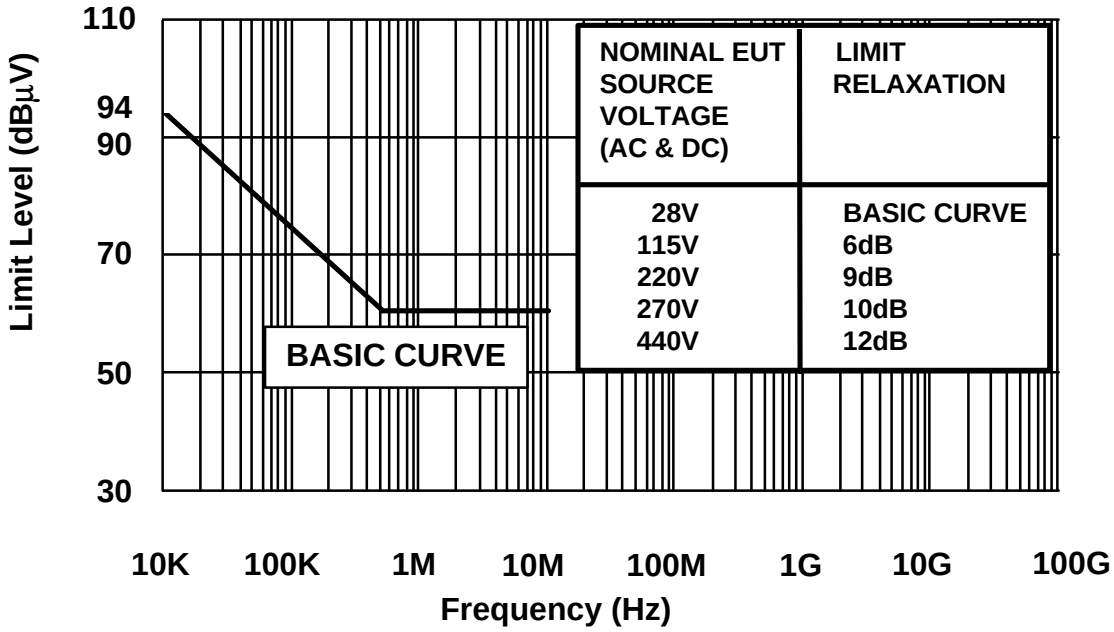


FIGURE 2. Conducted emissions.

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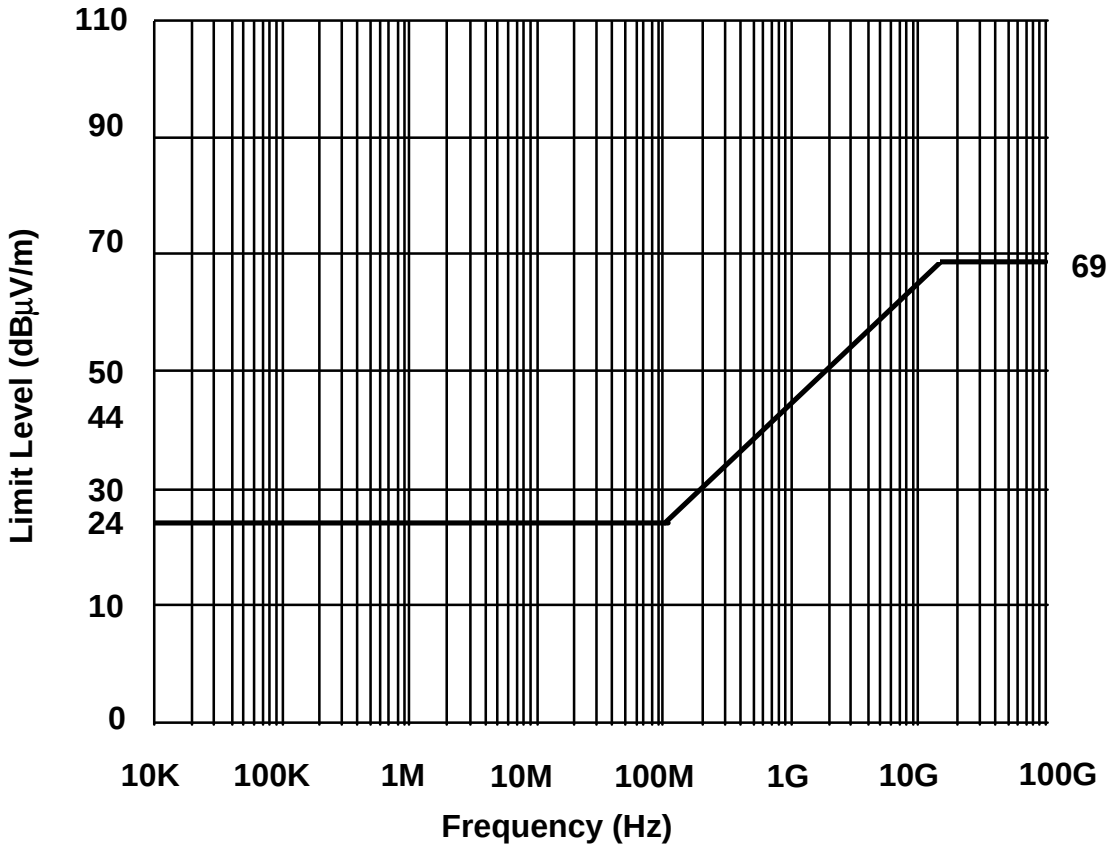
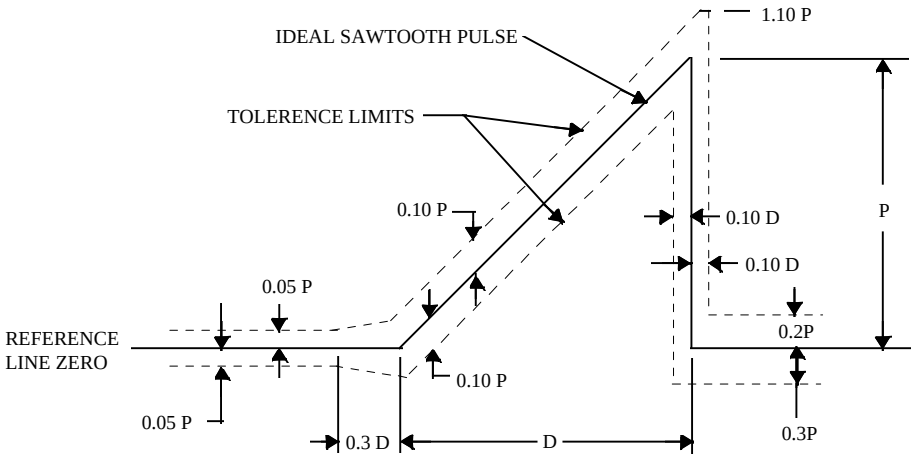


FIGURE 3. Radiated emissions.

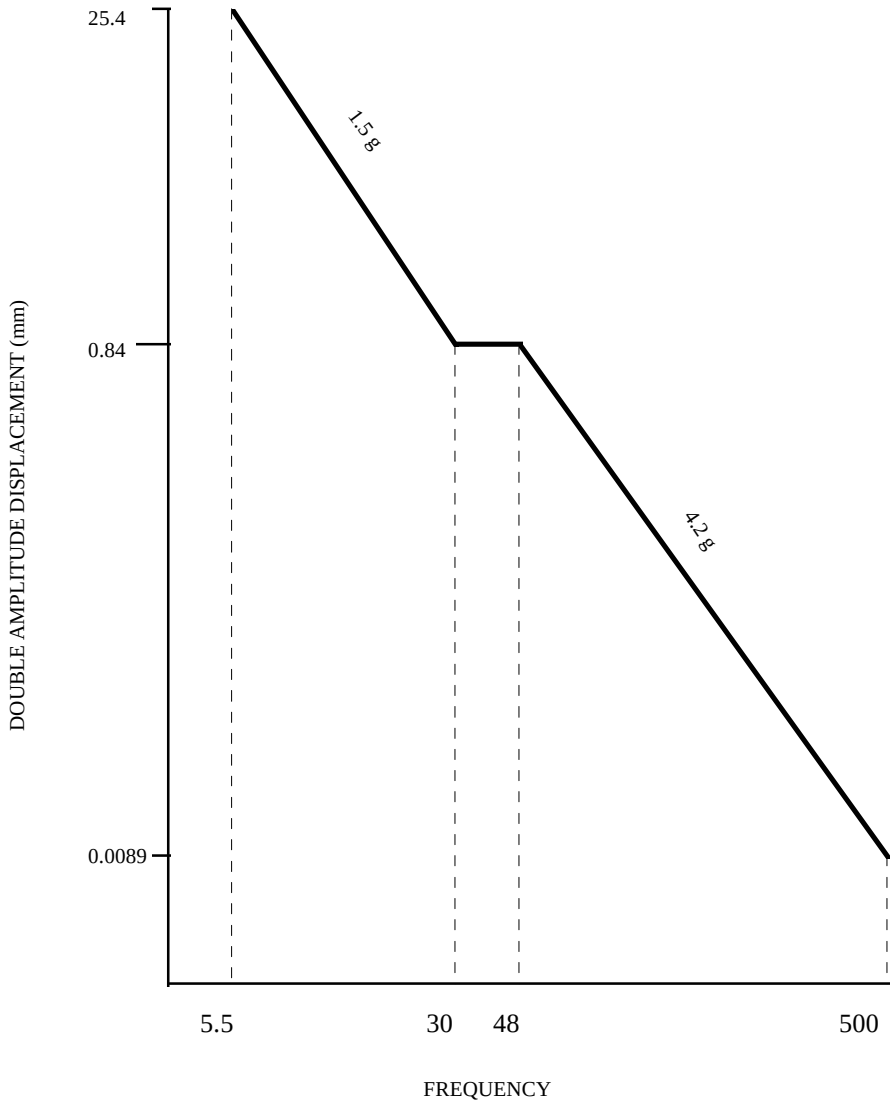
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NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of $V_i \pm 0.1\ V_i$ where V_i is the velocity-change associated with the ideal pulse which equals 0.5 PD. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 4. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

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NOTES:

1. Acceleration levels: ± 9 g.
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 5. Vibration curve.

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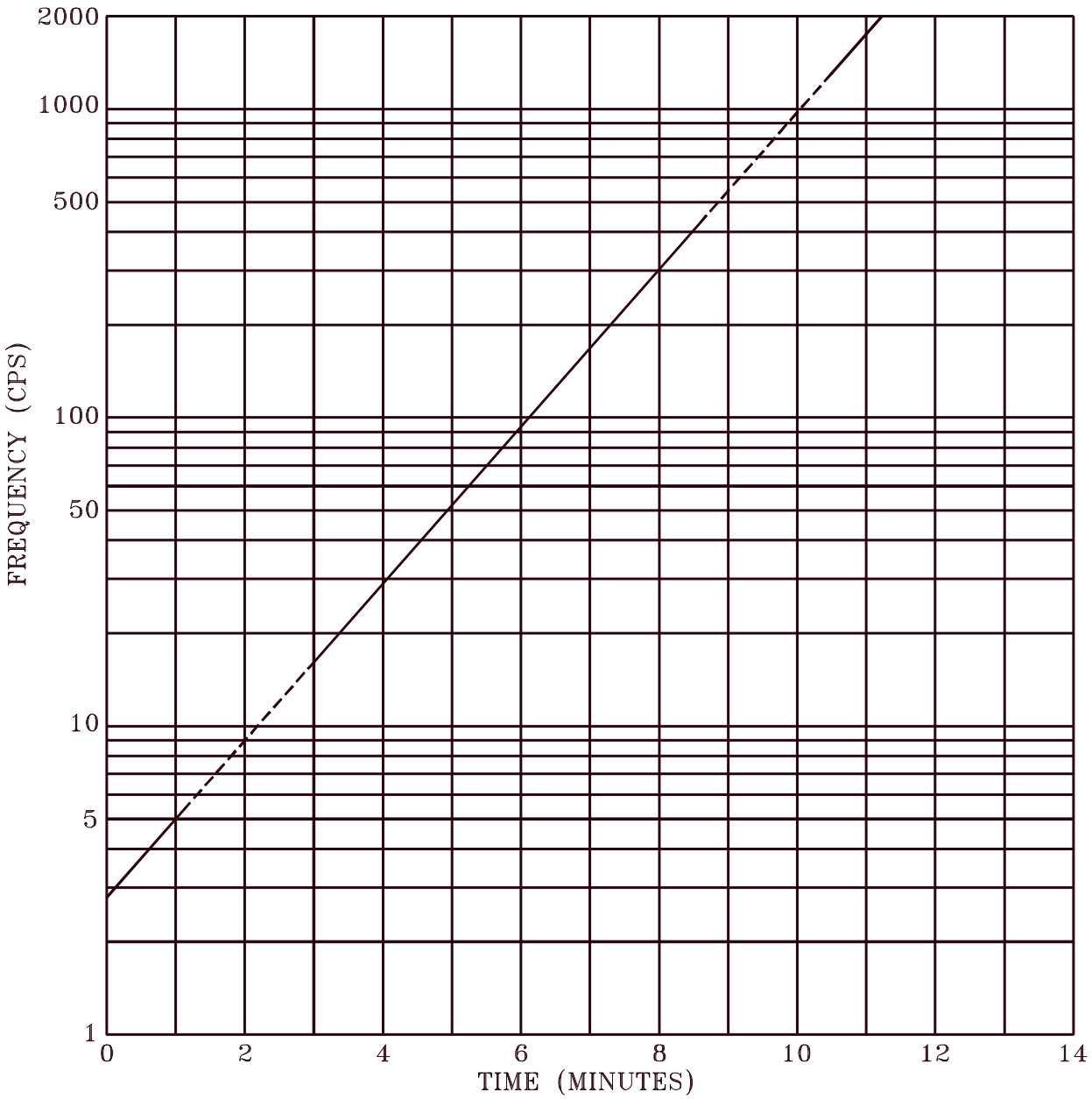


FIGURE 6. Logarithmic sweep.

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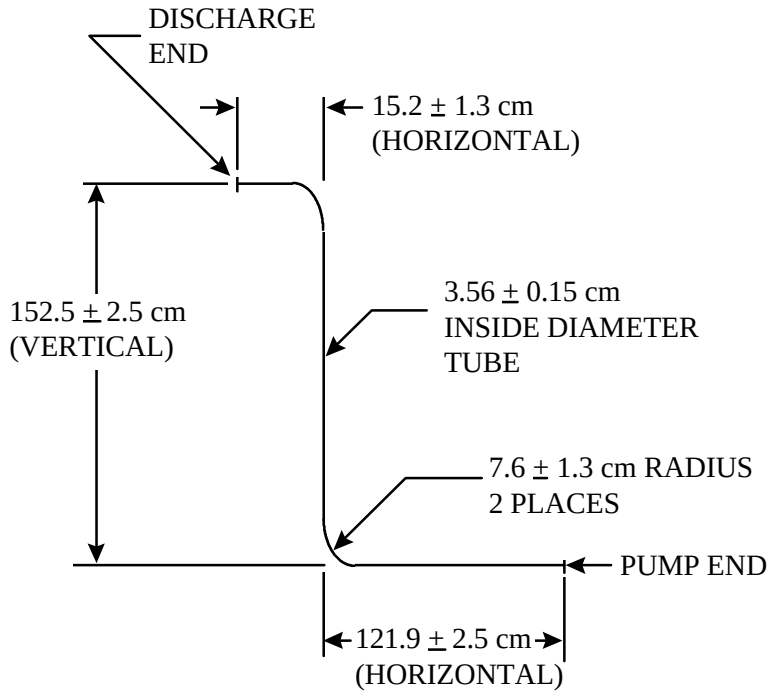


FIGURE 7. Discharge tube for dry-run test.

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(Project 4320-0056)

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3. DOCUMENT TITLE PUMP, SUBMERSIBLE, BILGE, ELECTRIC-MOTOR-DRIVEN

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

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c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	(810) 574-8745 786-8745 IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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